

EpistemoBrain: A Seven-Scientist Council with Dual-Graph Evidence Governance for Auditable Scientific Blindspot Discovery

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Abstract

Scientific literature review is among the highest-value uses of large language model (LLM) agents, and also among the riskiest. A fluent, well-cited summary can hide a causal claim built on cross-sectional correlation, six papers that draw on a single dataset, or a dissenting view that the summarizer quietly averaged away. We present **EpistemoBrain**, an agentic framework in which **seven AI scientists with distinct specializations** review a contested empirical question under a formal council protocol (independent precommitment, professional acquisition, evidence exchange, cross-examination, blindspot bounty, joint modeling, and final rejudgment), and produce an *auditable controversy evidence map*: conclusions sit next to their limitations, every claim traces to its source, and no dissent is deleted.

EpistemoBrain treats memory as the public cognitive layer of a scientific community rather than a private per-agent cache. Three layers, private process memory, collective executive memory, and a formal evidence graph, are organized as a *dual-graph* architecture: the process graph records how the scientists worked, while the evidence graph records what the community now knows. Every formal change passes through an append-only Scientific Event Ledger, a six-stage Evidence Gate, and a single Graph Projector whose exclusive write privilege is enforced at the database level. The protocol never uses majority voting to settle scientific questions. Outcomes take the form of conditional consensus, and genuine disagreement persists as named Dissent Certificates.

We evaluate EpistemoBrain on the **Foresight-Blindspot** benchmark, a time-sliced evaluation protocol for blindspot discovery in computational social science. The study comprises a controlled scripted-case comparison of five baselines (single-agent deep research, fixed multi-agent debate, a council with linear context, a council with executive memory but no evidence engine, and the full system) and a

six-variant ablation ladder that removes one capability at a time. Blindspot coverage follows seat coverage: a lone researcher misses six of seven specialist blindspots. Removing the adversarial falsifier or the evidence auditor each costs a gold blindspot, and removing lineage tracking inflates the system’s own evidence-independence report. A real-model study further shows that keyword-based blindspot recall fails on free-form text and that a semantic judge repairs the measurement. Across repeated runs we find a stable precision–recall trade-off rather than sampling noise: the full system’s governance attains perfect blindspot precision at the price of the lowest recall (two of seven gold concepts), every simplified variant, including the lone researcher, finds nearly all gold concepts at lower precision, and removing precommitment restores recall 1.0 in every run. End-to-end verification on real research tasks is reported alongside.

1 Introduction

LLM agents have made deep research practical: a user asks a question, and an agent retrieves, reads, and synthesizes the literature into a coherent answer (Yao et al., 2023; Lewis et al., 2020). For contested scientific questions, the standard pipeline carries three failure modes that a fluency-optimized summary cannot repair. A single agent that reads a paper tends to see only its own interpretation; the measurement scientist’s concern about self-report bias is invisible to a pipeline that has no measurement scientist. Agreement is not evidence either. If seven agents read the same erroneous abstract, the same retracted paper, or the same misattributed citation, the system can package a *shared error* as a consensus that looks verified simply because it was repeated (Ioannidis, 2005; Open Science Collaboration, 2015). Finally, a summarizer’s closing “everyone basically agrees” step erases exactly the information a contested review must preserve: what the dissenting scientist opposed, and which

evidence would change their mind. These are not prompt-tuning problems. They are structural questions about how a scientific review should be organized.

Computational social science makes the stakes concrete. The reproducibility crisis has shown that a large share of published findings do not replicate (Open Science Collaboration, 2015), that publication bias inflates apparent effect sizes (Fanelli, 2010), and that researcher degrees of freedom multiply false positives (Gelman and Loken, 2013; Nosek et al., 2018). In the domain this work targets, digital behavior, social media, and mental health, cross-sectional designs are routinely reported as causal, self-report measures dominate, and a small number of datasets support a large fraction of the literature (Orben and Przybylski, 2019; Valkenburg et al., 2022). An evidence reviewer for this domain needs more than an accurate summary. It has to *audit* claims against their sources, *count independent evidence* rather than papers, and *keep the dissent* visible.

We propose **EpistemoBrain**, the organizing brain of **Poliscope**. It is a deep research agent that assembles **seven AI scientists**: theory builder, causal scientist, measurement and construct scientist, statistics and replication scientist, boundary and context scientist, adversarial falsifier, and evidence and provenance auditor. The scientists interact through a seven-round council protocol. The design starts from the opposite direction of “seven chatbots plus a summarizer”: we first decide what organizational structure and memory mechanisms a reliable controversy review requires, and then place the roles inside that structure. Four pillars carry the architecture. First, all seven seats participate in every task; cost is controlled through speaking budgets, semantic deduplication, and shared retrieval caches rather than by dropping seats. Second, dissent is a first-class object. Majority voting is forbidden, outcomes are conditional consensus, and dissents become named certificates. Third, process and evidence live in separate graphs, coupled by an append-only event ledger and a single database-enforced writer. Fourth, memory is a layered public cognitive layer (private per-seat memory, collective executive memory, and the formal evidence graph), with MemoBrain’s native operations redefined in scientific semantics as Quarantine, Dialectical Fold, and Perspective Recall.

We evaluate EpistemoBrain on **Foresight-Blindspot**, a time-sliced evaluation protocol for

blindspot discovery in computational social science. The evidence comes from a controlled comparison of five baselines and a six-variant ablation ladder on a scripted case with no model calls, no network, and no database, together with end-to-end verification on real research tasks. Blindspot coverage follows seat coverage: after we corrected the single-agent baseline’s seat-selection semantics (Section 4.2), it finds one of seven gold blindspots, while every seven-seat variant finds all seven. The ablations locate the seats that carry coverage (removing the adversarial falsifier or the evidence auditor each costs a gold blindspot) and expose a self-report trap: removing lineage inflates the system’s own evidence-independence score. Under scripted gateways, the remaining mechanisms (memory, evidence gate) are not observable on the blindspot metric, and we identify this as a boundary of the measurement rather than a claim about the mechanisms.

Our contributions are threefold: (1) We introduce a *council protocol* for scientific controversy review. Seven specialized scientist agents interact through structured scientific actions (propose, support, challenge, qualify, fork, request, revise, dissent) under an explicit prohibition on majority voting, producing conditional consensus with minority positions preserved. (2) We design *dual-graph evidence governance*: a process graph and an evidence graph coupled by an append-only, idempotent, replayable Scientific Event Ledger, a six-stage Evidence Gate, and a single Graph Projector whose write privileges are enforced at the database privilege level. A layered memory model redefines executive-memory operations in scientific semantics. (3) We release **Poliscope**, an open-source implementation, together with the **ForesightBlindspot** evaluation protocol, a five-baseline comparison, and end-to-end verification, establishing a measurement basis for blindspot discovery in computational social science.

2 Related Work

2.1 Multi-Agent Debate and Collaboration

Multi-agent debate improves reasoning by having agents critique one another’s answers (Du et al., 2023; Liang et al., 2023; Chen et al., 2023; Qian et al., 2024), and role-specialized panels have been proposed for domain reasoning. These systems typically close with a summarization step that aggregates positions, and none addresses the failure

mode we consider central: the summarizer can flatten a genuine dissent, and the agents can converge on a shared evidence error that looks like independent verification. EpistemoBrain differs on both axes. It forbids majority voting as a truth mechanism, and it subjects evidence, rather than arguments, to a six-stage gate before anything enters the formal knowledge graph.

2.2 Memory for Long-Horizon Agents

Long-horizon reasoning strains bounded contexts, which motivates explicit memory architectures (Packer et al., 2023; Sumers et al., 2023). MemoBrain introduces executive memory for tool-augmented agents: a dependency-aware graph of reasoning steps with Flush, Fold, and Recall operations that prune invalid steps, fold completed sub-trajectories, and preserve a compact, high-salience reasoning backbone under a fixed context budget (Qian et al., 2026). EpistemoBrain adopts MemoBrain as its executive-memory foundation but changes the unit of analysis. MemoBrain manages *one* agent’s trajectory; EpistemoBrain manages a *community*’s shared cognitive state. Three redefinitions follow. Flush becomes Quarantine: a refuted claim changes status instead of disappearing. Fold becomes Dialectical Fold: a dispute compresses only after both strongest positions, the hinge variables, and the falsification conditions are preserved. Recall becomes Perspective Recall: each seat recalls a role-specific projection of the shared evidence graph. Process memory and formal evidence are additionally separated into two graphs, because a scientist’s momentary guess must never quietly upgrade into a formal conclusion.

2.3 Retrieval-Augmented Generation and Factual Verification

RAG grounds generation in retrieved documents (Lewis et al., 2020), and verification pipelines check claims against evidence (Thorne et al., 2018; Wadden et al., 2020; Min et al., 2023; Dhuliawala et al., 2023). These systems verify *textual entailment*. Auditing a *scientific claim* requires checks that standard entailment systems do not perform: evidence levels (full text with exact quotes versus abstract-only metadata), citation entailment (an introductory hypothesis must not be reported as a tested result), causal-upgrade policy (cross-sectional correlation must not become a causal claim), evidence independence (papers sharing a dataset count as one independent evidence cluster), and source prove-

nance (retraction status, preprint lineage). We reuse the production rules for these audits in our scoring functions rather than re-deriving parallel approximations.

2.4 LLM Agents for Science

LLM agents are increasingly applied to scientific tasks, from hypothesis generation to literature analysis. A recurring concern is that agentic systems can produce authoritative-sounding claims without grounding (Ioannidis, 2005; Open Science Collaboration, 2015). Our response is to make grounding a governance property rather than a prompt property. We are not aware of a system that combines a seven-scientist council protocol, dual-graph governance, an append-only event ledger with database-enforced write isolation, and a time-sliced blindspot evaluation benchmark in one open-source implementation. To our knowledge, Poliscope is the first.

3 Method

3.1 Overview and Formalization

Figure 1 gives the architecture. A researcher submits a Research Contract: the question, scope, budget, and an optional knowledge base and set of PDFs. The system proposes candidate claims, and the researcher confirms which ones to investigate. A worker then runs the seven-seat council through its seven deliberative rounds (eight phases when the fixed human checkpoint is counted), appending every formal change as an event to the Scientific Event Ledger. The Graph Projector, the ledger’s only consumer and the evidence graph’s only writer, applies the six-stage Evidence Gate to each candidate and materializes the admitted nodes and edges. Gaps in the evidence graph generate blindspot bounties, which are scored, ranked, and broadcast back into the council (epistemic routing). The researcher finishes with a Research Brief, an interactive controversy map, an audit trail, and a synthesized final paper.

Following the formalization style of executive-memory models, we make the data flow explicit. A task is a contract $X = (q, s, b, K)$ over a question q , scope s , budget b , and optional materials K . At step t a seat issues a structured action a_t ; every formal change is encoded as an event

$$e_t = (t, \text{actor}, \text{type}, \text{payload}, \text{refs}), \quad (1)$$

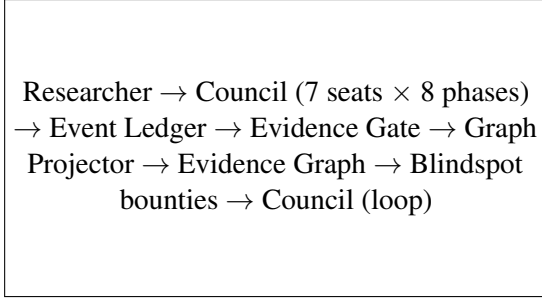


Figure 1: EpistemoBrain architecture: council, ledger, gate, projector, and epistemic routing form a closed loop. Process memory (left, per-seat MemoBrain) and the formal evidence graph (right) are physically separated.

and the ledger grows only by admission:

$$L_{t+1} = \begin{cases} L_t \oplus e_t, & \text{Gate}(e_t) = \text{admit}, \\ L_t, & \text{otherwise,} \end{cases} \quad (2)$$

where $\oplus$ is idempotent append keyed by a deterministic idempotency key. The evidence graph is a pure projection of the admitted ledger,

$$G = \text{Project}(L), \quad (3)$$

so replaying the ledger on an empty database reconstructs G exactly, and no other code path can write it. Each seat’s working context is a role-specific sum of process recall and an evidence projection,

$$C_i = \text{ProcessRecall}_i(L) + \text{EvidenceProjection}_i(G, \text{role}_i). \quad (4)$$

which is why two seats never read the same undifferentiated summary. A gap u exposed by G becomes a bounty scored on five dimensions,

$$\begin{aligned} b(u) = & w_1 \text{impact}(u) + w_2 \text{uncertainty}(u) \\ & + w_3 \text{investigability}(u) + w_4 \text{novelty}(u) \\ & - w_5 \widehat{\text{cost}}(u). \end{aligned} \quad (5)$$

and the final round either produces a conditional consensus

$$C^* = \text{JointModel}(\{C_i\}_{i=1}^7, G), \quad (6)$$

or refuses when the strongest opposing position or a falsification condition is missing. Equations (2)–(3) encode the write isolation of Section 3.4, Eq. (5) drives the routing of Section 3.5, and Eq. (6) is the no-voting rule of Section 3.2 made formal.

Seat	Responsibility
Theory Builder	mechanisms, competing explanations
Causal Scientist	confounding, reverse causation, selection
Measurement Scientist	operationalization, self-report bias
Replication Scientist	power, sensitivity, replication history
Boundary Scientist	population, culture, platform limits
Adversarial Falsifier	strongest counterexamples
Evidence Auditor	provenance, entailment, independence

Table 1: The seven scientist seats and their primary responsibilities.

3.2 The Seven-Scientist Council

Each research task runs all seven seats. Table 1 lists the seats, the question each owns, and its primary responsibilities. Every seat has an independent role specification, independent private state and private memory, and a different evidence-projection weighting. Seats gather evidence without seeing one another and commit positions first; evidence is exchanged only afterward, so cross-examination examines genuinely independent positions rather than restating a shared draft.

Scientists submit only structured actions: PROPOSE, SUPPORT, CHALLENGE, QUALIFY, FORK, REQUEST, REVISE, DISSENT; when challenged they may DEFEND, REVISE, NARROW, WITHDRAW, or DISSENT. Every action carries an actor, a target, a statement, evidence references, a confidence, a falsification condition, and a novelty mark. The ledger rejects factual assertions without evidence, target-less comments, duplicated positions, and agreement that adds no information.

The protocol runs seven rounds. (1) *Independent Precommitment*: seats commit initial judgments and confidence levels without seeing one another (sealed first, read later). (2) *Professional Acquisition*: seats issue evidence requests from their own perspectives; a query planner merges them and executes against shared sources (OpenAlex, Crossref, Semantic Scholar, Unpaywall, and user-provided materials). (3) *Evidence Exchange*: structured evidence is exchanged, never private reasoning histories. (4) *Cross-Examination*: targeted challenges on causality, measurement, statistics, boundaries, sources, and hidden assumptions. (5) *Blindspot Bounty*: gaps exposed by the evidence graph are scored on impact, uncertainty, investigability, nov-

elty, and cost, then ranked and claimed. (6) *Joint Modeling*: a conditional consensus is drafted together with the shared understanding, the strongest supporting and opposing positions, hinge variables, scope boundaries, unresolved conflicts, and falsification conditions. (7) *Final Rejudgment*: seats judge independently again, recording belief updates and named dissents.

No majority vote decides scientific truth. The joint-modeling round refuses to produce a consensus when the strongest opposing position or a falsification condition is missing. A seat that cannot accept the consensus receives a Dissent Certificate, a first-class node in the evidence graph that records what it opposes, why, and what evidence would retract the dissent.

3.3 Layered Memory and Dialectical Operations

Memory is organized in three layers. *Private memory* holds each scientist’s tool calls, failed routes, private hypotheses, and plans. *Collective executive memory* records the community’s structured cognitive events: who proposed what, who attacked whom, which controversies remain open, and which blindspots nobody has claimed. *The evidence graph* holds the audited formal knowledge: atomic claims, study findings, sources, constructs, contexts, blindspots, debate capsules, and dissent certificates.

MemoBrain’s native operations are re-defined for the scientific setting. *Quarantine* (from Flush) keeps a refuted claim instead of deleting it: the claim moves through PROPOSED → SUPPORTED → CONTESTED → QUARANTINED → RESURRECTED/REJECTED, and its record keeps who refuted it, what evidence is missing, and which conditions would resurrect it. *Dialectical Fold* (from Fold) compresses a dispute into a Debate Capsule only after preserving the shared understanding, the strongest supporting evidence, the strongest opposing evidence, hinge variables, scope boundaries, unresolved conflicts, and falsification conditions; the original nodes remain, and the capsule links back to them. *Perspective Recall* (from Recall) builds seat *i*’s context as in Eq. (4): a role-specific slice of the shared evidence graph added to that seat’s process recall, so the seven seats never read one undifferentiated summary. Three dynamics complete the set. *Fork* opens parallel research paths when evidence cannot

distinguish competing explanations. *Resurrect* re-activates a quarantined hypothesis when new evidence meets its resurrection conditions. *Merge* records conditional merge candidates for the researcher to adjudicate.

3.4 Dual-Graph Governance

Process and conclusion are stored apart. The *Process Graph* answers “how did the scientists work”: tasks, tool calls, failed routes, challenges, and decisions, managed by MemoBrain and freely foldable. The *Evidence Graph* answers “what do we now know”: ten node types (ResearchQuestion, Claim, Source, StudyFinding, Construct, Context, Blindspot, DebateCapsule, DiscriminatingStudy, DissentCertificate) and twelve edge types (SUPPORTS, REFUTES, QUALIFIES, CONTRADICTS, CONFOUNDS, MEDIATES, MODERATES, OPERATIONALIZES, DERIVED_FROM, APPLIES_IN, EXPOSES, TESTS).

The two graphs are coupled by the *Scientific Event Ledger*. Every formal change is first appended as an idempotent, replayable, auditable event, and the *Graph Projector*, the ledger’s only consumer and the evidence graph’s only writer, applies the Evidence Gate and materializes the admitted nodes and edges (Eqs. (2)–(3)). Write isolation is enforced at the database privilege level, not by coding discipline. The application role holds no insert, update, or truncate privilege on the evidence-graph tables, and the projector role holds no privilege on the ledger, so a bug on either side cannot forge the other’s data. Refuted, quarantined, and folded nodes are never physically deleted; they change status and remain traceable.

3.5 The Evidence Gate

Every candidate event passes six stages: *schema validation*, *semantic deduplication*, *source verification*, *citation entailment*, *method quality*, and *graph consistency*. Evidence is graded A through D. Level A requires full text with exact quotes and precise location; Level B is abstract plus meta-data; Level C covers second-hand descriptions in reviews; Level D covers web or news leads. Levels B, C, and D can never alone support a high-confidence causal conclusion. Three audits apply to formal findings: source authenticity (DOI, title, authors, version, retraction, PDF identity), citation entailment (whether the quote supports the claim, whether correlation is reported as causation, whether qualifiers were dropped), and method

quality (directness, design strength, measurement validity, statistical precision, replicability, external validity). A *CausalUpgradePolicy* intercepts the combination of a cross-sectional design with a causal claim at stage six and quarantines the event with an audit record. The pipeline also tracks evidence independence. Lineage detection classifies same-dataset, overlapping-sample, preprint, extension, reanalysis, and meta-analysis-inclusion dependencies, and the interface reports both the paper count and the independent-evidence-cluster count: six papers sharing one dataset count as one independent evidence cluster.

Four mechanisms defend against the subtler failure of *shared* evidence errors. *Blind evidence review* strips prestige signals (author, journal, citation counts) from first substantive judgments. *Independent dual extraction* extracts high-impact findings twice through separate paths and escalates discrepancies. *Source-diversity constraints* generate a Blindspot node automatically when all supporting evidence for a claim shares a dataset, team, country, or design. *Adversarial retrieval* runs six classes of reverse queries (refutation, null results, alternative theories, measurement criticism, replication failure, boundary inversion) against the evidence sources for each confirmed claim.

3.6 Epistemic Routing and Blindspot Bounty

The council is not a scripted play-through of seven rounds. Gaps exposed by the evidence graph generate blindspot bounties (Eq. (5)), which are scored, ranked, and broadcast for seats to claim. The evidence state, rather than a moderator reading from a script, drives the next investigation direction. A blindspot is consequently a first-class object in the product (a radar view ranked by impact and investigability), not a limitations paragraph. When a bounty round closes, the system may also propose a *DiscriminatingStudy*: a design with competing predictions that would resolve an open conflict, together with its expected information gain.

3.7 Human Directional Checkpoint

After the blindspot bounty and before joint modeling, the task pauses at one fixed checkpoint (AWAITING_COUNCIL_INPUT). The researcher may review the seven seats’ precommitment positions and submit a short directional note (for example, “prioritize cross-cultural boundary conditions”), or explicitly leave it blank and continue; both paths are equally valid. The note enters joint

modeling only under the explicit label “researcher directional note, not a scientific judgment”. It never changes an Evidence Gate decision, never serves as evidence for or against a claim, and never enters a Debate Capsule or Dissent Certificate. Identical evidence input must produce identical claim admission, dissents, and gate decisions with or without a note, a property asserted by tests. The checkpoint is directional control, not an eighth vote.

3.8 Engineering Realization

Poliscope is a modular monolith with an asynchronous worker and PostgreSQL as the source of truth; no workflow state lives only in memory. The worker claims tasks with `SELECT ... FOR UPDATE SKIP LOCKED`. A council run executes each phase in one transaction as the application role and commits; the projector then runs in a second transaction as the projector role. Two identities, two transactions, one ledger. All model calls go through a Model Gateway (OpenAI-compatible endpoints, streaming with a non-streaming fallback, per-task model configuration, cost and latency accounting), and all external tools go through a Tool Gateway (OpenAlex, Crossref, Semantic Scholar, Unpaywall, user knowledge bases). A single-seat failure degrades the round: the seat is marked absent with a reason on the ledger instead of aborting the task. Structured-output repair failures are quarantined and never write the formal graph. Live views are served as two SSE streams, the formal ledger stream and a best-effort process stream for token-level progress, so the audit trail is never polluted by reasoning noise. The implementation is open-source (MIT) at <https://github.com/Fishman-free/poliscope>, archived at <https://doi.org/10.5281/zenodo.22541350>.

4 Experiments

4.1 The ForesightBlindspot Benchmark and Baselines

We design **ForesightBlindspot**, a time-sliced evaluation protocol for blindspot discovery in computational social science (domain: digital behavior, social media, and mental health). Each case is a contested question with a cutoff date: the system may read only the pre-cutoff corpus. Gold blindspots are gaps that later high-quality research confirmed, that had reasonable premonitory evi-

dence before the cutoff, and that do not depend on entirely unforeseeable platforms, events, or methods. Leakage controls are built into the protocol: closed pre-cutoff corpora, no future citations, anonymizable variables, and an explicit disclosure that base-model parameters may contain future knowledge, a risk that cannot be fully eliminated. The metric family covers blindspots (recall, precision, high-impact recall, specificity, actionability, foresight distance), evidence (citation existence, citation entailment, evidence independence, causal-overclaim rate, full-text coverage), memory (compression ratio, scientific-backbone retention, contradiction retention, long-horizon drift, recovery quality), collaboration (role diversity, debate utility, belief-update quality, dissent preservation, false-consensus rate), and cost (cost per valid blindspot, tokens per admitted finding).

The five baselines are configurations of the real council rather than separate implementations, so each adds exactly one capability over the previous: (1) *Single-Agent Deep Research*, one researcher (the theory builder) with no protocol, no memory, and no gate; (2) *Fixed Multi-Agent Debate*, seven seats with an undifferentiated deliberator, no per-seat memory, and no gate; (3) *Council + Linear Context*, seven seats sharing one undifferentiated transcript; (4) *Council + MemoBrain without Evidence Engine*, seven seats with per-seat memory but the gate removed; and (5) *Full Poliscopes*, the complete system.

4.2 Baseline Construction Correctness

Before reporting numbers, we validated that the baselines are what they claim to be, and the first evaluation run exposed two construction defects. First, the single-agent baseline selected its seat as the alphabetically first seat of an ordered enumeration, which happened to be the adversarial falsifier. The “lone researcher” was in fact the falsifier, and in our scripted material the falsifier holds the strongest blindspot answers, so the single agent scored as high as the full council (F1 0.8 on all five variants). Second, the final-rejudgment handler iterated all seven seats unconditionally, minting placeholder judgments for seats that never participated. We corrected both: the single-agent baseline explicitly uses the theory builder, and the rejudgment handler respects the participating-seat set with a compatible fallback that leaves the seven-seat production path byte-for-byte unchanged. A regression guard now asserts that single-seat blindspot

Baseline	Rec.	Prec.	F1
Single-Agent Deep Research	0.143	0.500	0.222
Fixed Multi-Agent Debate	1.000	0.778	0.875
Council + Linear Context	1.000	0.778	0.875
Council + MemoBrain, no gate	1.000	0.778	0.875
Full Poliscopes	1.000	0.778	0.875

Table 2: Blindspot recall, precision, and F1 on the scripted demo case (B_dev). All four seven-seat variants find all seven gold blindspots; the single agent, restricted to the theory builder, finds one.

coverage is strictly below multi-seat coverage, so a future regression that collapses the baseline ladder fails immediately. We report this correction because the construction of the measurement instrument is part of the measurement.

4.3 Main Results on the Scripted Case

Table 2 reports the five baselines on a scripted demo case (screen time and adolescent depressive symptoms) in which every gateway is deterministic: no model calls, no network, no database. Each seat nominates one specialist blindspot in the bounty round (the falsifier nominates two, one deliberately duplicating the boundary scientist’s nomination to model overlapping multi-seat nominations). Gold blindspots are one keyword per specialist seat.

Three findings stand out. First, blindspot coverage is driven by *seat coverage*. The lone researcher finds one of seven specialist blindspots (F1 0.222), while every seven-seat variant finds all seven (F1 0.875). The council’s contribution is real and large, and it is attributable to the set of seats rather than to the presence of any single one. Second, what matters is the *set of participating seats*, not a particular seat: after the construction fix, the single agent’s score is determined by what its own seat contributes, not by whether the falsifier happens to be present. Third, the four seven-seat variants score identically on the blindspot metric under scripted gateways, and this is a structural consequence of the harness. Every variant collects all seven seats’ outputs, the scripted gateway ignores prompts and memory, and bounty events are admitted unconditionally by the gate. The contributions of the protocol, memory, and evidence gate to blindspot *integration* are therefore not observable through this metric on scripted material. They surface in the auxiliary scores (dissent preservation, citation entailment, evidence independence) and require real-model runs to observe on blindspots. We treat this as an identified boundary of the measurement,

not as a claim about the mechanisms.

4.4 Ablation Study

Beyond the five-baseline ladder, we ablate the full system one capability at a time (design spec 11.4): six *full minus X* variants, each removing exactly one mechanism while everything else, including the evidence gate, stays intact. The removals are (1) independent precommitment (the phase does not run); (2) the adversarial falsifier seat; (3) the evidence auditor seat; (4) the dialectical fold (JOINT MODELING skips the DebateCapsule and records an unfilled slot instead of silently dropping the opposition); (5) evidence lineage (acquired sources arrive without dataset metadata, so evidence independence cannot see that two papers share one dataset); (6) MemoBrain (per-seat private memory is replaced by one undifferentiated transcript). Each ablation is a configuration of the same orchestrator the full system runs. The switch is one flag or one seat subset, not a reimplementaion, so the contrast with Full Poliscope isolates that capability’s contribution.

Table 3 shows three mechanism-level effects. Removing the adversarial falsifier drops recall from 1.000 to 0.857: the publication-bias blindspot was its nomination, and no other seat covers it, so a council without a dedicated falsifier misses exactly the gap that role exists to find. Precision rises at the same time (0.778 to 0.857), because the falsifier’s second nomination deliberately duplicates the boundary scientist’s, and without the falsifier no statement is double-paid. Removing the evidence auditor also costs recall (1.000 to 0.857, F1 0.800): the provenance blindspot belonged to that seat. Removing lineage raises evidence independence from 0.667 to 1.000. Without dataset metadata, two papers that share a cohort are counted as independent evidence, so the number the system reports about its own evidence quality is *inflated by the absence of the very mechanism that would correct it*. The dialectical-fold ablation cannot move the blindspot metric, but it is visible in the honesty signal: the DebateCapsule disappears and the run records one additional unfilled slot rather than pretending the debate had no opposition. Dissent preservation is a property of the protocol, and its absence is recorded, not hidden.

Two ablations show no effect on this scripted material, and we say so plainly rather than manufacturing a difference. Removing precommitment removes the seven PRECOMMITMENT_SEALED

events but leaves every score unchanged; removing MemoBrain changes nothing either. This is the same structural boundary identified for the baseline ladder: a scripted gateway answers identically regardless of prompt, memory, or phase history, so protocol-level integration effects (what precommitment changes about later reasoning, what per-seat memory changes about what a seat recalls) are not observable without real models. The three ablations that do show effects (falsifier, auditor, lineage) do so because they change a *participant* with scripted content or the data a scoring function sees, not because the mechanism’s behavioral consequences are being measured. We therefore read the three effects as validated mechanism checks and the two null results as an identified measurement boundary, not as evidence that the two capabilities contribute nothing.

4.5 Real-Model Evaluation and the Semantic Judge

The scripted case is deterministic, and its blindspot metric rests on a *keyword* proxy: “reverse_causation” matches only statements that literally contain those words (underscore-separated and case-folded). The scripted gateway writes its statements to contain the keyword, so the proxy works there. It does not survive contact with a real model. Free-form output such as “the direction of effect could run the other way” names the reverse-causation gap without containing the token, so the keyword score reports recall 0.0 even when the gap was found. To measure blindspot quality under a real model we added a *semantic judge*: for each gold concept, the same model gateway (no second vendor SDK) is asked which numbered candidate statements semantically address that concept, and recall and precision are derived with the identical definitions the keyword score uses. A judge that is unreachable, or whose structured answer is quarantined, returns *unmeasured* rather than silently 0.0.

We then ran five key variants (the full system, the single-agent baseline, and the three mechanism ablations whose scripted effects were largest) through the real gateway (DeepSeek-V4-Flash on the official DeepSeek API), three runs each, and scored blindspots with the semantic judge. An initial eleven-variant single-run sweep was abandoned after the first endpoint rate-limited for two days and polluted twelve runs (unfilled slots jumped to 46, and the judge was unreachable for every se-

Variant	Rec.	Prec.	F1	Indep.	Unfilled
Full Poliscopescope	1.000	0.778	0.875	0.667	1
– Precommitment	1.000	0.778	0.875	0.667	1
– Adversarial falsifier	0.857	0.857	0.857	0.667	1
– Evidence auditor	0.857	0.750	0.800	0.667	1
– Dialectical fold	1.000	0.778	0.875	0.667	2
– Lineage	1.000	0.778	0.875	1.000	1
– MemoBrain	1.000	0.778	0.875	0.667	1

Table 3: Ablations of the full system on the same scripted case (B_dev). “Indep.” is evidence independence; “Unfilled” counts slots the run records as unfinished rather than fabricating. Each ablation removes exactly one capability; the adversarial falsifier and the evidence auditor each carry a gold blindspot of their own, which is why removing them costs recall.

semantic call); those runs were discarded rather than reported. Table 4 reports the three-run means (F1 is recomputed from mean recall and precision, not averaged over per-run F1s).

Three findings follow. *First*, the semantic judge repairs the measurement: the keyword proxy reports recall 0.0 for the full system on free-form text, while the semantic judge produces a readable, non-degenerate distribution. *Second*, repeated runs turn the earlier single-run reading (“sampling variance dominates”) into a stable, reproducible *precision–recall trade-off*. Full Poliscopescope achieves perfect precision (1.000) at the cost of the lowest recall (0.286, exactly two of seven gold concepts in all three runs). Every simplified variant, including the lone researcher, finds nearly all gold concepts (recall 0.809–1.000) at markedly lower precision (0.406–0.586). Removing precommitment reproduces recall 1.000 in every run, so the earlier single-run observation was a real, repeated effect rather than a sampling artifact. Unfilled slots explain the gap: the full system leaves 12.3 slots unfilled on average (the model produced no structured nomination), against 1.0–1.7 for the simplified variants. Protocol burden on a flash-tier model is the binding constraint: governance buys precision at the cost of recall, and under a weak model the recall cost dominates F1. *Third*, the deterministic data-level scores (citation entailment 0.5, evidence independence 0.667, dissent preservation 1.0) are identical across all fifteen runs, as expected for scoring that does not depend on model output.

4.6 Analysis: What the Trade-off Says About Deployment

The real-model results are not a negative result, and reading them as one would miss the paper’s central design claim. The governance mechanisms EpistemoBrain adds operate at the *integration* level: they

shape what a seat reasons and recalls across rounds, and their benefit needs a model strong enough to sustain the full protocol without the seat absence observed here (12.3 unfilled slots against 1.0–1.7). What the repeated runs establish is a deployment map. When evidence quality is the priority (a contested review where an admitted causal claim can be wrong in a costly way), the full protocol is the only configuration with perfect blindspot precision and intact honesty signals: dissent preservation 1.0 across all fifteen runs, citation entailment and evidence independence computed from data rather than asserted. When coverage is the priority and the available model is a flash-tier one, a lighter protocol (or precommitment removed) recovers recall 0.809–1.000 at a known, quantified precision cost. The scripted case isolates each mechanism’s structural contribution; the real-model runs locate the model-strength boundary at which that isolation stops. Both pieces are needed to choose a configuration honestly, and the system reports whichever side of the trade-off it is running on instead of averaging the two into a single score.

4.7 System Verification

Beyond the controlled comparison, Poliscopescope is verified end to end. The test suite comprises 752 unit tests and 364 integration tests over a real PostgreSQL instance with three database identities, including privilege tests that assert the application role cannot write the evidence graph and the projector role cannot rewrite the ledger: dual-graph isolation is a database fact, not a code convention. Real research tasks run through the full seven-seat council on real OpenAI-compatible endpoints, producing conditional consensus, Debate Capsules, Dissent Certificates, and a synthesized final paper whose references are bound to DOIs. Tasks that exhaust budget or lose seats finish as “completed

Variant	Sem-R	Sem-P	Sem-F1	Unf.	n
Single-Agent Deep Research	0.809	0.544	0.651	1.0	3
Full Poliscopescope	0.286	1.000	0.444	12.3	3
– Precommitment	1.000	0.490	0.658	1.0	3
– Adversarial falsifier	1.000	0.406	0.577	1.7	3
– Evidence auditor	0.952	0.586	0.726	1.3	3

Table 4: Blindspot scoring on repeated real-model runs, means over $n = 3$. “Sem” is the semantic judge; R/P/F1 are recall, precision, and F1; “Unf.” is the mean number of unfinished evidence slots. The keyword proxy is omitted: on this endpoint it again collapses to 0 for the full system, and the semantic judge is the controlled measurement.

with gaps”, with every absent seat and unfilled slot named in the report rather than fabricated into a success. The event ledger is idempotent and replay-safe (re-running a task is a no-op), and the streaming live view survives reconnection via Last-Event-ID.

5 Conclusion

We presented EpistemoBrain, the organizing brain of the open-source Poliscopescope system: a seven-scientist council with a formal protocol, layered memory, dual-graph evidence governance, an append-only event ledger with database-enforced write isolation, an evidence gate with three-layer audit, and epistemic routing through blindspot bounties. The design thesis is that reliable blindspot discovery is an organizational and governance problem rather than a prompt problem. Dissent has to survive the pipeline, evidence has to be audited before it becomes a conclusion, and a process note has to stay distinguishable from evidence. The ForesightBlindspot protocol, the five-baseline comparison, and the six-variant ablation ladder establish a measurement basis, with one open caveat: scripted gateways make the protocol’s integration-level contributions unobservable on the blindspot metric, which is exactly why two of the six ablations are null on this material. The real-model study shows that a keyword proxy for blindspot recall fails on free-form text and that a semantic judge repairs the measurement; repeated runs then expose a stable deployment boundary rather than noise. Under a flash-tier model, full governance buys perfect blindspot precision at the cost of recall (every simplified variant, including a lone researcher, finds more gold concepts), and seat absence (12.3 unfilled slots against 1.0–1.7) is the binding constraint. Observing the integration-level benefits therefore requires either a stronger model or a deliberately lighter protocol, and the system is designed to report which regime it is in.

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References

- Shunyu Yao, Jeffrey Zhao, Dian Yu, Nan Du, Izhak Shafran, Karthik Narasimhan, and Yuan Cao. 2023. ReAct: Synergizing reasoning and acting in language models. In *International Conference on Learning Representations (ICLR)*.
- Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich K  ttler, Mike Lewis, Wen-tau Yih, Tim Rock-t  schel, et al. 2020. Retrieval-augmented generation for knowledge-intensive NLP tasks. In *Advances in Neural Information Processing Systems (NeurIPS)*.
- Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, Brian Ichter, Fei Xia, Ed Chi, Quoc Le, and Denny Zhou. 2022. Chain-of-thought prompting elicits reasoning in large language models. In *Advances in Neural Information Processing Systems (NeurIPS)*.
- Shunyu Yao, Dian Yu, Jeffrey Zhao, Izhak Shafran, Thomas Griffiths, Yuan Cao, and Karthik Narasimhan. 2023. Tree of thoughts: Deliberate problem solving with large language models. In *Advances in Neural Information Processing Systems (NeurIPS)*.
- Noah Shinn, Federico Cassano, Edward Gopinath, Karthik Narasimhan, and Shunyu Yao. 2023. Reflexion: Language agents with verbal reinforcement learning. In *Advances in Neural Information Processing Systems (NeurIPS)*.
- Aman Madaan, Niket Tandon, Prakhar Gupta, Skyler Hallinan, Luyu Gao, Sarah Wiegreffe, Uri Alon, Nouha Dziri, Shrimai Prabhumoye, Yiming Yang, et al. 2023. Self-refine: Iterative refinement with self-feedback. In *Advances in Neural Information Processing Systems (NeurIPS)*.

- Yilun Du, Shuang Li, Antonio Torralba, Joshua B. Tenenbaum, and Igor Mordatch. 2023. Improving factuality and reasoning in language models through multiagent debate. *arXiv preprint arXiv:2305.14325*.
- Tian Liang, Zhiwei He, Wenxiang Jiao, Xing Wang, Yan Wang, Rui Wang, Yujiu Yang, Zhaopeng Tu, and Shuming Shi. 2023. Encouraging divergent thinking in large language models through multi-agent debate. *arXiv preprint arXiv:2305.19118*.
- Weize Chen, Yusheng Su, Jingwei Zhan, Xiaoyi Zhang, Yujia Chen, Songtao Yuan, Juntong Chan, Xinchuan Yu, Zhangzhi Zhou, et al. 2023. AgentVerse: Facilitating multi-agent collaboration and exploring emergent behaviors. *arXiv preprint arXiv:2308.10848*.
- Chen Qian, Wei Liu, Hongzhang Yang, Bingcong Cong, Shenghui Li, Zhiyuan Liu, Mingjie Xiong, Ge Wu, Haoqi Wang, et al. 2024. ChatDev: Communicative agents for software development. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (ACL)*.
- Charles Packer, Sarah Wooders, Kevin Lin, Vivian Fang, Shishir Patel, and Joseph E. Gonzalez. 2023. MemGPT: Towards LLMs as operating systems. *arXiv preprint arXiv:2310.08560*.
- Theodore R. Sumers, Shunyu Yao, Karthik Narasimhan, and Thomas L. Griffiths. 2023. Cognitive architectures for language agents. *arXiv preprint arXiv:2309.02427*.
- Hongjin Qian, Zhao Cao, and Zheng Liu. 2026. Mem-oBrain: Executive memory as an agentic brain for reasoning. In *Findings of the Association for Computational Linguistics: ACL 2026*.
- James Thorne, Andreas Vlachos, Christos Christodoulopoulos, and Arpit Mittal. 2018. FEVER: A large-scale dataset for fact extraction and VERification. In *Proceedings of the 2018 Conference of the North American Chapter of the Association for Computational Linguistics (NAACL-HLT)*.
- David Wadden, Shanchuan Lin, Kyle Lo, Lucy Lu, Madeleine Yoshioka, Hanjie Wei, Arman Cohan, Iz Beltagy, Kenton Lee, and Noah A. Ostendorf. 2020. Fact or fiction: Verifying scientific claims. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*.
- Sewon Min, Kalpesh Krishna, Xinxi Lyu, Mike Lewis, Wen-tau Yih, Pang Wei Koh, Mohit Iyyer, Luke Zettlemoyer, and Hannaneh Hajishirzi. 2023. FActScore: Fine-grained atomic evaluation of factual precision in long form text generation. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing (EMNLP)*.
- Shehzaad Dhuliawala, Mojtaba Komeili, Jing Xu, Roberta Raileanu, Xian Li, Asli Celikyilmaz, and Jason Weston. 2023. Chain-of-verification reduces hallucination in large language models. *arXiv preprint arXiv:2309.11495*.
- Tu Vu, Mohit Iyyer, Xuezhi Wang, Noah Constant, Jerry Wei, Yuancheng Xu, Jeffrey Flanigan, Daniel Hsu, Omar Albalak, Mark Yatskar, et al. 2023. FreshLLMs: Refreshing large language models with search engine augmentation. *arXiv preprint arXiv:2310.03214*.
- John P. A. Ioannidis. 2005. Why most published research findings are false. *PLoS Medicine*, 2(8):e124.
- Open Science Collaboration. 2015. Estimating the reproducibility of psychological science. *Science*, 349(6251):aac4716.
- Andrew Gelman and Eric Loken. 2013. The garden of forking paths: Why multiple comparisons can be a problem, even when there is no “fishing expedition” or “p-hacking” and the research hypothesis was posited ahead of time. *Department of Statistics, Columbia University*.
- Brian A. Nosek, Charles R. Ebersole, Alexander C. DeHaven, and David T. Mellor. 2018. The preregistration revolution. *Proceedings of the National Academy of Sciences*, 115(11):2600–2607.
- Daniele Fanelli. 2010. “Positive” results increase down the hierarchy of the sciences. *PLoS ONE*, 5(4):e10068.
- Amy Orben and Andrew K. Przybylski. 2019. The association between adolescent well-being and digital technology use. *Nature Human Behaviour*, 3(2):173–182.
- Patti M. Valkenburg, Adrian Meier, and Ine Beyens. 2022. Social media use and its impact on adolescent mental health: An umbrella review of the evidence. *Current Opinion in Psychology*, 44:58–68.
- Jason Priem, Heather Piwowar, and Richard Orr. 2022. OpenAlex: A fully-open index of scholarly works, authors, venues, institutions, and concepts. *arXiv preprint arXiv:2205.01833*.
- Uri Simonsohn, Leif D. Nelson, and Joseph P. Simmons. 2014. P-curve: A key to the file-drawer. *Journal of Experimental Psychology: General*, 143(2):534–547.

A Appendix: Implementation Notes

Deterministic evaluation harness. The Foresight-Blindspot harness (`packages/evaluation/`) runs the five baselines and the six ablation variants as configurations of the real orchestrator against a database-less event sink, so a run is deterministic and fast enough for unit tests; scoring functions reuse the production audit rules (causal upgrade policy, citation entailment verifier, lineage detection and evidence clustering) instead of re-deriving them, so a variant’s score reflects the same standard the gated system is held to. Each ablation is one protocol switch on that same orchestrator – a phase

subset, a seat subset, a fold flag, or a lineage-free acquirer – never a reimplementaion (Section 4.4). Gold blindspot matching is implemented as a keyword proxy, with a double-annotation agreement protocol (Kappa/Alpha statistics, adjudication) in place.

Permissions as design. PostgreSQL grants implement the write isolation: the application role can append to the ledger and read the evidence graph but cannot insert, update, or truncate graph tables; the projector role can write the graph and advance event status but cannot rewrite event payloads; neither role can truncate anything. A documented exception grants the application role DELETE for the session-lifecycle operation of destroying a whole task at the researcher’s explicit request – the one path by which evidence rows may be physically removed.

Reproducibility. The repository pins migrations (Alembic), seeds, and Docker Compose services (postgres, migrate, api, worker, web, caddy); the demonstration task seed runs the real orchestrator with scripted model and tool stand-ins, and the evaluation harness runs with scripted gateways, so all numbers in this paper are reproducible without external credentials.